An Exponential Lower Bound for Scarf's Algorithm on a Unit-Capacity Bipartite Matching Polytope

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Abstract

Faenza, He and Sethuraman asked whether Scarf's algorithm admits a polynomially bounded execution on the bipartite matching polytope for an arbitrary ordinal matrix C. The question is delicate because the unit-capacity matching polytope is highly degenerate, and a degeneracy rule might in principle shortcut a long nondegenerate path. This note gives a negative answer. For every k we construct a bipartite graph consisting of k disjoint edges and one isolated vertex, with the usual slack variables and right hand side $b = \mathbf{1}$, and an ordinal matrix C_k, such that every terminating execution of Scarf's algorithm from the standard initial pair has at least $2^k - 1$ iterations. There is also an execution with exactly $2^k - 1$ iterations. Since the instance has $2k + 1$ rows and $3k + 1$ columns, this is exponential in the input dimension. The ordinal matrix is a reflected Gray-code gadget. The main point is that duplicating each Gray-code row on the two endpoints of a matching edge removes the perturbation used in a nondegenerate construction while preserving a rank invariant: every cardinal pivot can increase the reflected-Gray rank of the current matching by at most one.

1 Introduction

Scarf's algorithm is a pivoting method for finding a dominating basis of an input triple (A, b, C), where $A = (I \mid A')$ is nonnegative, $b > 0$, and C is an ordinal matrix. Faenza, He and Sethuraman proved polynomial convergence for the stable-marriage choice of the ordinal matrix C^* on the bipartite matching polytope, using a carefully chosen rule for degenerate cardinal pivots [1]. They also asked whether polynomial convergence continues to hold for arbitrary ordinal matrices C on the same polytope.

The construction below shows that the answer is no, even in the strict unit-capacity case $b = \mathbf{1}$. The graph is as simple as possible: k independent matching edges, plus one isolated vertex. The degeneracy of the unit-capacity polytope does create many extra cardinal pivots, but none of them can jump forward by more than one step in a reflected Gray code. Thus the shortest terminating Scarf execution is still exponential.

I use the following standard convention. The first n columns of A are identity/slack columns. An ordinal matrix has

$$c_{i,i} = 0 < c_{i,e} < c_{i,j} \qquad (j \neq i,\ e \text{ a nonidentity column}),$$

and an n-set D is ordinal when, for $u_i = \min_{j \in D} c_{i,j}$, every nonidentity column e is dominated by at least one row:

$$\exists i \quad u_i \geq c_{i,e}.$$

Only the relative order in each row matters; larger entries are better.

2 The instance

Fix $k \geq 1$. The bipartite graph G_k has vertices

$$z_0, \quad a_1, b_1, \ldots, a_k, b_k,$$

where z_0 is isolated and $e_i = a_i b_i$ is the only non-slack edge incident with a_i, b_i. The rows of A_k are ordered as

$$z_0, a_1, \ldots, a_k, b_1, \ldots, b_k.$$

Let the identity/slack columns be

$$s_0, \quad r_i \text{ at } a_i, \quad t_i \text{ at } b_i \qquad (i = 1, \ldots, k),$$

and let the k nonidentity columns be e_i, with ones in rows a_i, b_i. Thus

$$A_k = (I \mid M_k), \qquad b = \mathbf{1},$$

where M_k is the node-edge incidence matrix of G_k. The number of rows is $2k+1$, and the total number of columns is $3k+1$.

We now define the ordinal matrix. First define permutations $\pi_i^{(k)}$ of $[k]$, for rows $i = 0, 1, \ldots, k$. The order is from worst to best. For $k = 1$,

$$\pi_0^{(1)} = \pi_1^{(1)} = (1).$$

For $k \geq 2$,

$$\begin{aligned} \pi_0^{(k)} &= (k, \pi_0^{(k-1)}), \\ \pi_i^{(k)} &= (\pi_i^{(k-1)}, k) \qquad (1 \leq i < k), \\ \pi_k^{(k)} &= (\pi_0^{(k-1)}, k). \end{aligned}$$

Let $\mathrm{pos}_i^{(k)}(j)$ be the position of j in $\pi_i^{(k)}$, so larger position means better. Define C_k as follows:

$$c_{v,s_v} = 0 \quad \text{for every row } v,$$

all other slack columns in row v receive distinct values larger than k, and edge columns receive

$$c_{z_0,e_j} = \mathrm{pos}_0^{(k)}(j),$$

$$c_{a_i,e_j} = c_{b_i,e_j} = \mathrm{pos}_i^{(k)}(j) \qquad (i, j \in [k]).$$

This is an ordinal matrix. Notice that the two endpoint rows a_i, b_i have identical preferences over edge columns, but different own slack columns.

For $k = 3$, the edge orders are

row	edge order, worst to best
z_0	e_3, e_2, e_1
a_1, b_1	e_1, e_2, e_3
a_2, b_2	e_1, e_2, e_3
a_3, b_3	e_2, e_1, e_3.

3 A compressed Gray-code lemma

It is useful first to fold the two endpoint rows of each edge into one row. Consider the compressed column set

$$s_0, s_1, \dots, s_k, e_1, \dots, e_k$$

with rows $0, 1, \dots, k$, own slack value 0, other slacks very large, and edge values $c_{i,e_j} = \mathrm{pos}_i^{(k)}(j)$.

Let

$$\mathrm{G}_k(0), \mathrm{G}_k(1), \dots, \mathrm{G}_k(2^k - 1)$$

be the binary reflected Gray code, with bit 1 the least significant bit. Thus

$$\mathrm{G}_k(q) = q \oplus (q \gg 1).$$

For a bit vector $x \in \{0,1\}^k$, let $\mathrm{R}_k(x)$ be its Gray-code rank, i.e. $\mathrm{G}_k(\mathrm{R}_k(x)) = x$. The predecessor and successor of x in this path, when they exist, differ from x in one bit.

For $x \in \{0,1\}^k$, define

$$d_i(x) = \begin{cases} s_i, & x_i = 0, \\ e_i, & x_i = 1, \end{cases} \qquad \bar{d}_i(x) = \begin{cases} e_i, & x_i = 0, \\ s_i, & x_i = 1. \end{cases}$$

For $h \in [k]$, set

$$D_k(x, h) = \{d_1(x), \dots, d_k(x)\} \cup \{\bar{d}_h(x)\}.$$

This is the compressed analogue of an almost-feasible ordinal basis; it omits s_0 and contains exactly one extra column, the complement of bit h.

Lemma 1 (Gray-code characterization)**.** *The compressed set $D_k(x, h)$ is an ordinal basis if and only if toggling bit h moves x to one of its neighbors on the binary reflected Gray-code path. Equivalently,*

$$\left| \mathrm{R}_k(x \oplus 2^{h-1}) - \mathrm{R}_k(x) \right| = 1.$$

At the two endpoints of the Gray-code path only the existing neighbor is allowed.

Proof. The proof is by induction on k. We also use the following terminal observation. In the compressed system, the set

$$B_k(x) = \{s_0, d_1(x), \dots, d_k(x)\}$$

is ordinal if and only if $x = (0, \dots, 0, 1)$, the last word of G_k. Indeed, if $x_k = 0$, then e_k is best in every row $1, \dots, k-1$, row k is inactive, and no active row can dominate e_k. If $x_k = 1$ and some lower bit is also one, then every active row sees a selected lower edge below e_k, so again no active row dominates e_k. If $x = (0, \dots, 0, 1)$, row k has utility $\mathrm{pos}_k^{(k)}(k) = k$ and dominates all edges.

For the induction step, write $x = (y, x_k)$ with $y \in \{0,1\}^{k-1}$. The reflected Gray code satisfies

$$\mathrm{G}_k = 0\,\mathrm{G}_{k-1},\ 1\,\mathrm{G}_{k-1}^{\mathrm{rev}},$$

so the first half has $x_k = 0$ and the second half has $x_k = 1$.

Suppose first that $x_k = 0$ and $h < k$. Then e_k is not selected in $D_k(x, h)$. Row 0 has e_k in position 1, so e_k is automatically dominated. On lower edges, row 0 and rows $1, \dots, k-1$ induce exactly the $(k-1)$-bit construction, up to adding one to row 0's positions; this shift does not change any comparison. Hence $D_k(x, h)$ is ordinal exactly when $D_{k-1}(y, h)$ is ordinal. By induction this means that toggling h moves y to a neighboring word of G_{k-1}, hence toggling h moves x to a neighboring word in the first half of G_k.

Still with $x_k = 0$, take $h = k$. Now e_k is selected, so row 0 has utility 1 and can only account for e_k. The lower edges must be dominated using only rows $1, \ldots, k-1$ and the selected lower edges. By the terminal observation in dimension $k-1$, this happens exactly when y is the last word of G_{k-1}. This is precisely the last word of the first half of G_k, whose successor is obtained by toggling bit k.

Now suppose $x_k = 1$. If $h = k$, the complement s_k is present and row k is inactive. Row 0 again has utility 1 because e_k is selected, so the lower edges must be dominated by rows $1, \ldots, k-1$ with no lower complement. By the terminal observation this holds exactly when y is the last word of G_{k-1}. This is the first word of the second half of G_k, whose predecessor is obtained by toggling bit k.

Finally let $x_k = 1$ and $h < k$. The selected lower edge set is nonempty because the complement of bit h is a lower edge when $y_h = 0$, and otherwise e_h itself is selected. The edge e_k is last in every row $1, \ldots, k$ and therefore does not affect the minimum utility of any active lower row. Row k induces on lower edges the old row 0 order. Thus the lower-edge domination test is exactly the test for $D_{k-1}(y, h)$. By induction, toggling h moves y to a neighbor in G_{k-1}; since the second half of G_k traverses G_{k-1} in reverse order, this is again exactly adjacency in the full k-bit Gray path. □

4 Lifting back to the unit-capacity matching polytope

A feasible basis B of $A_k x = \mathbf{1}$ that is reachable in the construction has the following form. It contains s_0. For each i either

$$\{r_i, t_i\} \subset B, \qquad e_i \notin B,$$

or

$$e_i \in B, \qquad |B \cap \{r_i, t_i\}| = 1.$$

In the first case put $x_i(B) = 0$, and in the second put $x_i(B) = 1$. The retained slack in the second case may be either r_i or t_i; it is a purely degenerate orientation choice.

If B has bit vector x and $h \in [k]$, let $\eta_h(B)$ be the complement column:

$$\eta_h(B) = e_h \quad \text{if } x_h = 0,$$

and if $x_h = 1$, $\eta_h(B)$ is the one of r_h, t_h that is not in B. The corresponding lifted almost-feasible ordinal basis is

$$D(B, h) = B \setminus \{s_0\} \cup \{\eta_h(B)\}.$$

Lemma 2 (Folding). *For every lifted pair $(B, D(B, h))$, the set $D(B, h)$ is ordinal for C_k if and only if the compressed set $D_k(x(B), h)$ is ordinal.*

Proof. For each i, the rows a_i and b_i have the same order on edge columns. If the own slack of one of these rows is present in $D(B, h)$, then that row has utility 0 and cannot dominate any edge. If its own slack is absent, its utility is the minimum, in the row order $\pi_i^{(k)}$, of exactly the edge set appearing in the compressed set $D_k(x(B), h)$. Thus every active endpoint row is an exact copy of compressed row i, and inactive endpoint rows contribute nothing. Row z_0 is exactly compressed row 0. Therefore the edge-domination inequalities are identical in the lifted and compressed systems. □

Lemma 3 (Cardinal choices). *Let $(B, D(B,h))$ be a nonterminal lifted Scarf pair.*

(a) *If $x_h(B) = 0$, then the entering column is e_h, and the only feasible cardinal leaving columns are r_h and t_h. Either choice changes the bit vector to $x(B) \oplus 2^{h-1}$.*

(b) *If $x_h(B) = 1$, then the entering column is the missing slack among r_h, t_h. The only feasible cardinal leaving columns are e_h and the retained slack. Leaving e_h changes the bit vector to $x(B) \oplus 2^{h-1}$; leaving the retained slack leaves the bit vector unchanged and merely swaps the degenerate orientation of bit h.*

No cardinal pivot can remove s_0 before termination.

Proof. The graph is a disjoint union of the isolated row z_0 and k independent two-row blocks. In the block for bit h, the three columns are

$$r_h = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad t_h = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad e_h = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The feasible two-column bases in this block are exactly $\{r_h, t_h\}$, $\{r_h, e_h\}$ and $\{t_h, e_h\}$. The two assertions follow by inspecting which one-column replacement keeps one of these three bases. Since every entering column before termination has zero in row z_0, removing s_0 would leave row z_0 unsupported and the basis singular. □

Lemma 4 (Reachability invariant). *Starting from the standard initial pair, every nonterminal Scarf pair reached by any sequence of cardinal pivot choices has the lifted form $(B, D(B,h))$ for a unique bit h. If an ordinal pivot inserts s_0, then the algorithm terminates.*

Proof. The initial pair has this form, with $B_0 = \{s_0, r_1, t_1, \ldots, r_k, t_k\}$ and $h = 1$. Suppose $(B, D(B,h))$ is a nonterminal lifted pair. By Lemma 3, after the cardinal pivot the new feasible basis B' is again of lifted form and still contains s_0. Moreover, if the leaving column is ℓ, then

$$B' = B \setminus \{\ell\} \cup \{\eta_h(B)\}, \qquad D(B,h) \setminus \{\ell\} = B' \setminus \{s_0\}.$$

The ordinal pivot inserts one column j not in the old ordinal basis. If $j = s_0$, then the new ordinal basis equals B', so the algorithm terminates. If $j \neq s_0$, then $j \notin B'$ because every column of B' other than s_0 already lies in $D(B,h) \setminus \{\ell\}$. The columns outside a lifted basis B' are exactly the complements: for an off bit the missing column is e_i, and for an on bit the missing column is the absent one of r_i, t_i, together with no other non-slack or slack column except columns already in B'. Thus $j = \eta_{h'}(B')$ for a unique bit h', and the next pair is $(B', D(B', h'))$. □

5 The lower bound

Theorem 1. *For the unit-capacity bipartite matching instance $(A_k, \mathbf{1}, C_k)$ constructed above, every terminating execution of Scarf's algorithm from the standard initial pair has at least*

$$2^k - 1$$

iterations. There exists a terminating execution with exactly $2^k - 1$ iterations.

Proof. The standard initial feasible basis is

$$B_0 = \{s_0, r_1, t_1, \ldots, r_k, t_k\}.$$

The initial ordinal basis is obtained by deleting s_0 and inserting the edge most preferred by row z_0. Since row z_0 orders the edges as

$$e_k, e_{k-1}, \ldots, e_1$$

from worst to best, this edge is e_1. Hence B_0 has bit vector 0^k, and the initial almost-feasible ordinal basis is $D(B_0, 1)$.

By Lemma 4, every nonterminal pair in any execution is a lifted Scarf pair $(B, D(B, h))$. For such a pair, Lemma 2 implies that $D_k(x(B), h)$ is ordinal. Lemma 1 therefore implies that toggling bit h changes the Gray rank by exactly one:

$$\left| \mathrm{R}_k(x(B) \oplus 2^{h-1}) - \mathrm{R}_k(x(B)) \right| = 1.$$

By Lemma 3, a cardinal pivot either toggles this bit or only swaps a zero slack with the other zero slack in the same two-row block. Consequently one Scarf iteration can increase the Gray rank $\mathrm{R}_k(x(B))$ by at most one.

The initial rank is $\mathrm{R}_k(0^k) = 0$. If the algorithm terminates, then the terminal basis contains s_0 and is ordinal. Folding the terminal basis gives the compressed basis $B_k(x)$, so by the terminal observation in the proof of Lemma 1, necessarily

$$x = (0, \ldots, 0, 1) = \mathrm{G}_k(2^k - 1).$$

Thus the terminal Gray rank is $2^k - 1$. Since each iteration increases this rank by at most one, every terminating execution has length at least $2^k - 1$.

For the matching upper bound, choose the cardinal leaving column so that the bit is toggled in the forward Gray-code direction whenever the current ordinal basis offers the predecessor/successor choice. When $x_h = 0$ either of the two slacks may leave; when $x_h = 1$ choose e_h to leave. Lemmas 1–3 then give exactly the reflected Gray-code walk

$$\mathrm{G}_k(0), \mathrm{G}_k(1), \ldots, \mathrm{G}_k(2^k - 1),$$

followed by insertion of s_0 in the ordinal basis. This execution has exactly $2^k - 1$ iterations. □

Corollary 1. *There is no polynomial p such that every arbitrary ordinal matrix on a unit-capacity bipartite matching polytope admits a terminating Scarf execution of length at most $p(n + m)$.*

Proof. The instance above has $n = 2k + 1$ rows and $n + m = 3k + 1$ columns, while its shortest terminating Scarf execution has $2^k - 1 = 2^{(n+m-1)/3} - 1$ iterations. □

6 Example: $k = 3$

For $k = 3$, the forward execution visits the bit vectors

$$000 \to 100 \to 110 \to 010 \to 011 \to 111 \to 101 \to 001,$$

where bit 1 is written on the left in this display. Cardinal degeneracy allows orientation choices whenever an edge enters an unmatched two-vertex block, and it allows zero-slack swaps when a missing slack enters a matched block. These choices can move backward or pause in Gray rank, but by Lemma 1 they cannot jump forward. Thus even the shortest execution has seven iterations.

7 Discussion

The construction shows that the special stable-marriage matrix C^* in [1] is essential for the known polynomial convergence result. The obstruction is not caused by capacities greater than one or by a nonmatching polytope: the graph is bipartite, $b = \mathbf{1}$, and every non-slack column is an ordinary matching edge. Degeneracy does not save the algorithm, because the duplicated endpoint rows turn all possible cardinal degeneracy choices into moves that increase reflected-Gray rank by at most one.

References

[1] Y. Faenza, C. He, and J. Sethuraman. Scarf's algorithm and stable marriages. *Mathematics of Operations Research*, 2025. Preprint available as arXiv:2303.00791.

[2] H. E. Scarf. The core of an N person game. *Econometrica*, 35(1):50–69, 1967.